

STAT 514 Final Exam

Fall 2025

Total 8 pages, 8 problems, and 100 points. 120 minutes.

LastName _____ FirstName _____

1. (16 points) Fill in the blank with the best answer.

- (a) A 2_{IV}^{8-3} design has _____ runs and _____ factors of 2 levels each.
- (b) In a resolution IV design, a two-factor interaction is not aliased with _____, but is possibly aliased with _____.
- (c) For a 2_{III}^{9-5} design, we can estimate _____ factorial effects (besides the intercept). Each main effect is aliased with _____ other factorial effects.
- (d) The fundamental principles for factorial effects are _____, _____, and _____.
- (e) A central composite design using a 2_V^{5-1} design and 3 center points has total _____ runs.
- (f) A 3^2 full factorial design has _____ runs and _____ factors.
- (g) A Latin hypercube design with n runs in k factors has _____ point(s) in each of the n equally sized intervals of every one-dimensional marginal, so that each level is sampled _____ time(s) per factor.
- (h) A maximin Latin hypercube design chooses design points to maximize the _____ of all pairwise distances, whereas a minimax design chooses points to minimize the _____ distance from any candidate input to its nearest design point.

2. (12 points) How many degrees of freedom are associated with the residuals in each of the following circumstances?
- (a) A completely randomized design to study three treatments with 7, 8, and 9 replicates respectively.
 - (b) A randomized paired comparison of two treatments A and B , where there are $n_A = 15$ observations with A and $n_B = 15$ observations with B .
 - (c) A full 2^4 factorial design with three replicates. The model includes all factorial effects.
 - (d) An unreplicated 2^{8-2} factorial design with resolution V. The model includes all the main effects and two-factor interactions.
 - (e) A central composite design using a 2^{3-1} factorial design and 5 center runs. A standard second-order model is fitted.
 - (f) A full 3^2 factorial design with two replicates. The model includes all factorial effects.
3. (15 points) For each of the following scenarios, please (1) identify the most appropriate design and briefly explain your reasoning; (2) clearly indicate any blocking factors used; and (3) identify the sample size for each experiment.
- (a) (3 points) In a screening study for a new adhesive, a team has identified seven potentially important formulation factors, each of which can only practically be run at “low” or “high” levels. Because material and test time are expensive, the team can run at most 16 experimental trials but still wants information on which factors are active and on the most important interactions.

- (b) (3 points) A machining engineer has identified three quantitative factors, cutting speed, feed rate, and depth of cut, as important. There are indications of curvature and possible interactions, and the engineer wants to fit a second-order model to predict surface roughness and locate a minimum within a practical operating region.
- (c) (3 points) Climate scientists use a global climate model with about 12 uncertain physical parameters. They want to explore the effects of these parameters on long-term temperature predictions and build an emulator that is valid across a wide parameter range. Replication of input settings is considered wasteful because the model is deterministic. They want to explore 100 uniformly spaced levels for each parameter.
- (d) (3 points) A biologist wants to compare four fish diets. She has 40 small aquaria in the same controlled-environment room, and the fish and tanks are judged to be essentially homogeneous. There are no obvious sources of systematic heterogeneity among experimental units.
- (e) (3 points) A production engineer wants to compare three levels of a new hardening treatment for steel parts. The trials must be done over five production days, and it is known that day-to-day variation has a large effect on hardness.

4. (8 points) An experiment was conducted to determine the effects of *Cutting speed* and *Coolant type* on the surface roughness of a turned steel shaft. The response variable “surface roughness” (in μm) is numeric, with larger values indicating a rougher surface.

Independent variable *Cutting speed* has three levels: Low, Medium, and High; independent variable *Coolant type* also has three levels: Dry, Water-based, and Oil-based. A completely randomized 3×3 factorial design with 4 replicates at each treatment combination was run (i.e., all 3×3 treatment combinations, each replicated 4 times).

- (a) (2 points) Write down the two-factor ANOVA model with interaction for this experiment, and state the zero-sum constraints for the model parameters.

- (b) (6 points) The fitted two-factor model with interaction explains 94% of the total variation in surface roughness ($R^2 = 0.94$). A partially completed ANOVA table is given below. Fill in missing entries in the ANOVA table.

Source	df	SS	MS	F
Cutting speed				90
Coolant type		90		
Interaction				
Error		18		
Total				

5. (14 points) An experiment was performed to study the effect of three factors (A, B, C) on the surface roughness of a machined part. A single replicate of a 2^3 design was run. The results follow:

Run	A	B	C	y
1	–	–	–	98
2	+	–	–	72
3	–	+	–	87
4	+	+	–	85
5	–	–	+	99
6	+	–	+	79
7	–	+	+	87
8	+	+	+	80

- (a) (2 points) Estimate the main effect of A .
- (b) (2 points) Estimate the AB interaction effect.
- (c) (4 points) Draw (by hand) the A main effect plot and the AB interaction plot.
- (d) (3 points) The half-normal plot of the factorial effects (not shown) suggests that only A and AB are significant. Using coded variables $x_A, x_B \in \{-1, +1\}$, fit the reduced regression model

$$y = \beta_0 + \beta_A x_A + \beta_{AB} x_A x_B + \varepsilon.$$

Give their least-squares estimate for β_0 , β_A , and β_{AB} .

- (e) (3 points) Using the fitted model from part (d), obtain the predicted surface roughness for each combination of A and B . Which settings of A and B minimize the predicted surface roughness? What is the corresponding predicted roughness?

6. (10 points) Consider the following design

A	B	C	D	E
-1	-1	-1	-1	-1
1	-1	-1	1	1
-1	1	-1	1	-1
1	1	-1	-1	1
-1	-1	1	1	1
1	-1	1	-1	-1
-1	1	1	-1	1
1	1	1	1	-1

(a) (2 points) What is the generator for column D ?

(b) (2 points) What is the generator for column E ?

(c) (2 points) What is the complete defining relation?

(d) (2 points) What is the resolution of this design?

(e) (2 points) A second fraction is obtained by reversing the signs of column B . What is the complete defining relation for the second fraction?

7. (13 points) The following table shows the design matrix and response data for an experiment that studies the effect of two factors A and B on the response time (y) in a chemical process.

Run	A	B	y
1	-1	-1	54
2	-1	1	47
3	1	-1	32
4	1	1	45
5	0	0	40
6	0	0	38
7	0	0	42

- (a) (2 points) Estimate the main effect of A .
- (b) (2 points) Estimate the AB interaction.
- (c) (3 points) Estimate the variance σ^2 . What are the degrees of freedom for estimating the variance?
- (d) (3 points) Is the AB interaction significant at 5% level? Show your work.
- (e) (3 points) Is the curvature effect significant at 5% level? Show your work.

Here is part of the t table showing the upper α percentage points of the t distribution.

df	1	2	3	4	5	6	7	8	9	10
$\alpha = 0.05$	6.314	2.92	2.353	2.132	2.015	1.943	1.895	1.86	1.833	1.812
$\alpha = 0.025$	12.706	4.303	3.182	2.776	2.571	2.447	2.365	2.306	2.262	2.228

8. (12 points) An engineer studies the effect of two (coded) variables x_1 and x_2 on a response y using a central composite design with three center points. A second-order model (in coded units) is obtained:

$$y = 8 + 4x_1 - 2x_2 - 2x_1^2 - x_2^2 + \varepsilon.$$

Assume a circumscribed central composite design with axial distance $\alpha = \sqrt{2}$.

- (a) (3 points) Write down the central composite design explicitly by listing all design points (x_1, x_2) , clearly indicating the factorial points, the axial points, and the center points.
- (b) (3 points) The second-order model can be written in matrix form

$$y = \beta_0 + x'b + x'Bx + \varepsilon,$$

where $x = (x_1, x_2)'$. Write down the vector b and the matrix B explicitly.

- (c) (3 points) Find the stationary point of the response surface.
- (d) (3 points) Is the stationary point a maximum point, a minimum point, or a saddle point? Justify your answer using the eigenvalues of B . (Hint: for a diagonal matrix, the eigenvalues are its diagonal entries.)